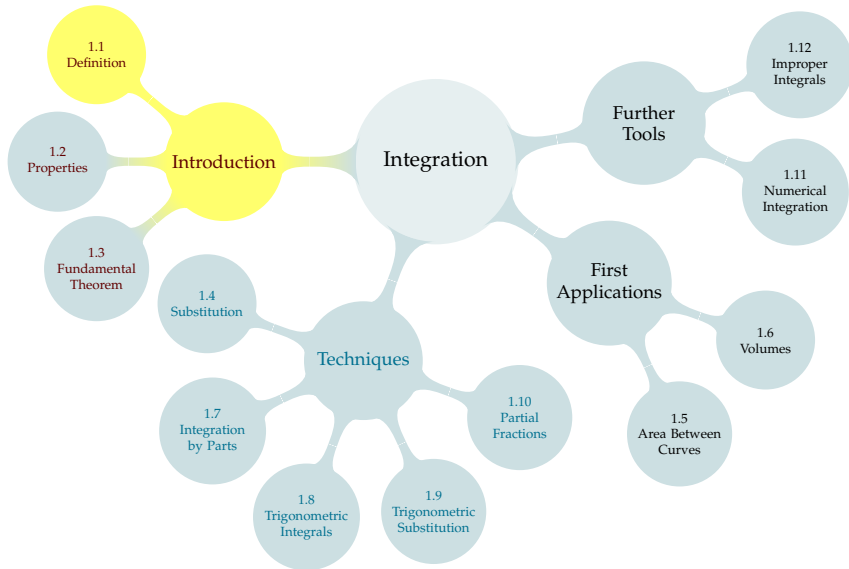


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We defined the definite integral as

$$\int_a^b f(x) \, dx = \lim_{N \rightarrow \infty} \sum_{i=1}^N \Delta x \cdot f(x_{i,N}^*)$$

where $\Delta x = \frac{b-a}{N}$ and $x_{i,N}^*$ is a point in the interval $[a + (i-1)\Delta x, a + i\Delta x]$.

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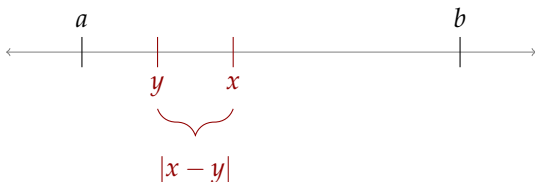
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We'll start with some general ideas that appear in the proof.

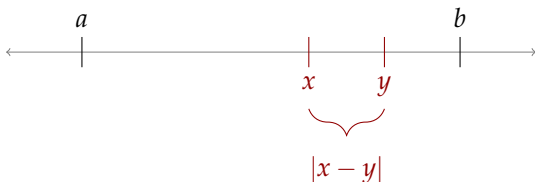
Suppose x and y are both in the interval $[a, b]$. What is the maximum possible value of $|x - y|$?



Proposition 1: distance between two numbers in an interval

If $a \leq x \leq b$ and $a \leq y \leq b$, then

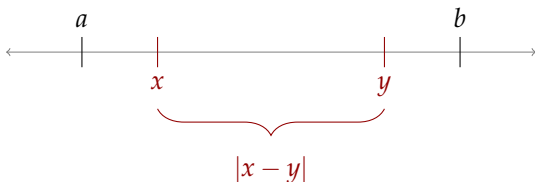
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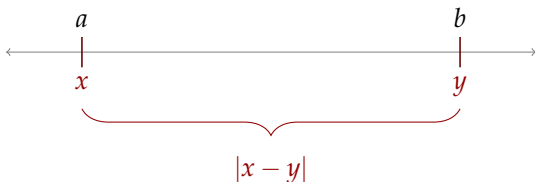
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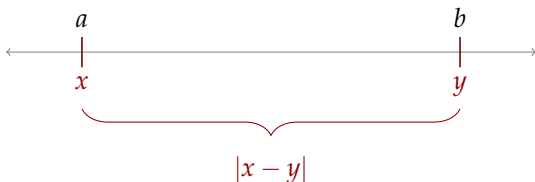
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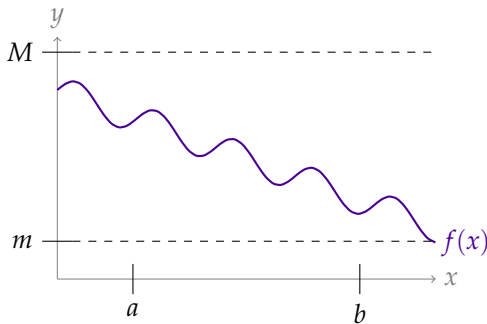


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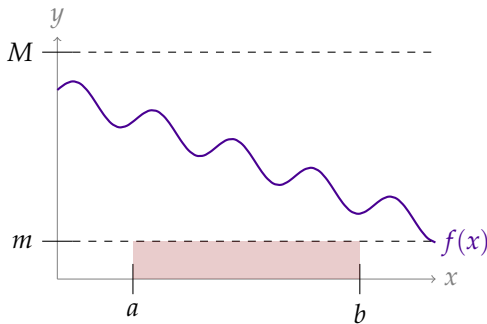
Proposition 2: area inequality

Let $f(x)$ be a function, defined over the interval $[a, b]$. If $m \leq f(x) \leq M$ over the entire interval $[a, b]$, then the (signed) area between the curve $y = f(x)$ and the x -axis, from a to b , is between $m(b - a)$ and $M(b - a)$.



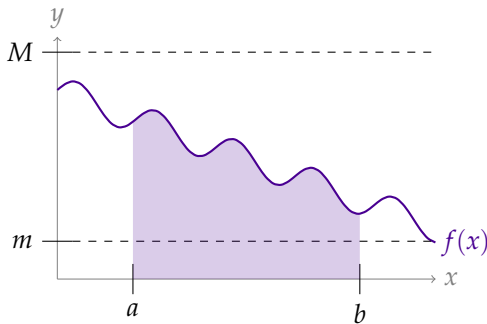
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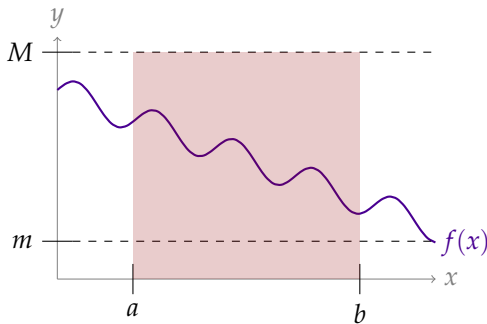
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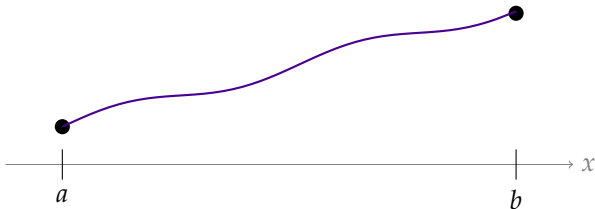


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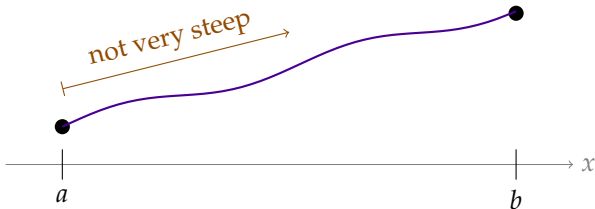
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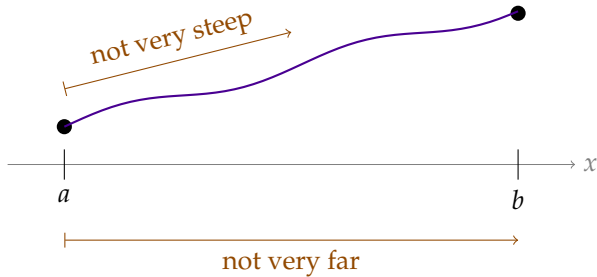
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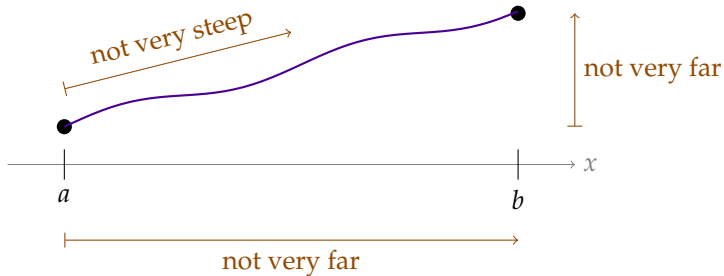
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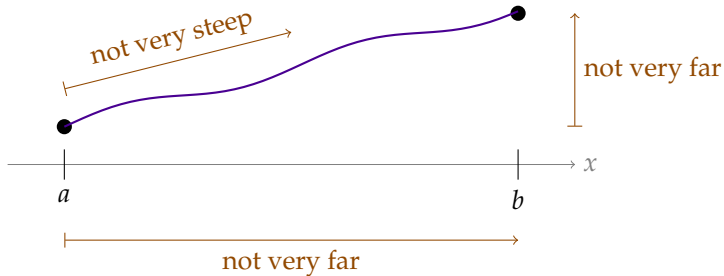
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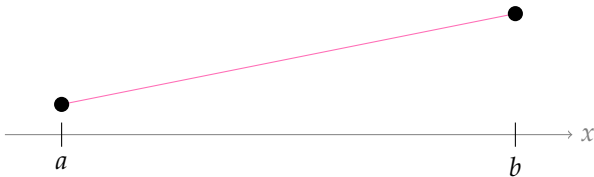
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The Mean Value Theorem provides a more explicit connection between these quantities.



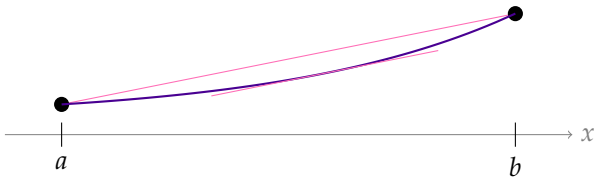
Mean Value Theorem

Let a and b be real numbers with $a < b$. Let f be a function such that

- ▶ $f(x)$ is continuous on the closed interval $a \leq x \leq b$, and
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Then there is a c in (a, b) such that

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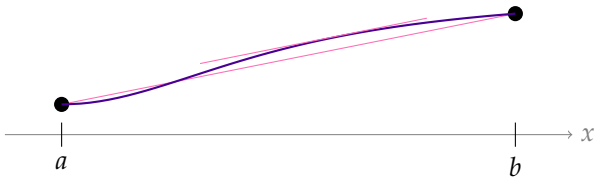
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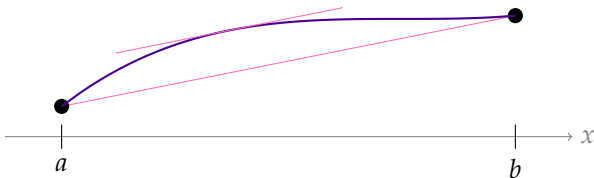
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Triangle Inequality

For any real numbers $x_1, x_2, \dots, x_n$:

$$\left| \sum_{i=1}^n x_i \right| \leq \sum_{i=1}^n |x_i|$$

Intuition: If some terms are positive and some are negative, they “cancel each other out” and make the overall sum smaller.

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$$|1 + 2|$$

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Proof outline:

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Proof outline:

Let x and y be any real numbers.

- ▶ $x \leq |x|$ and $y \leq |y|$, so $x + y \leq |x| + |y|$
- ▶ $-x \leq |x|$ and $-y \leq |y|$, so $-(x + y) = (-x) + (-y) \leq |x| + |y|$
- ▶ $|x + y| = \begin{cases} x + y & \text{if } x + y \geq 0 \\ -(x + y) & \text{if } x + y < 0 \end{cases} \leq |x| + |y|$
- ▶ Then $|x + y + z| = |(x + y) + z| \leq |x + y| + |z| \leq |x| + |y| + |z|$, etc.

REQUIREMENTS

We will consider

$$\int_a^b f(x) \, dx$$

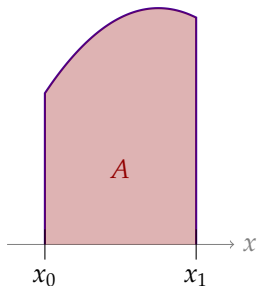
where:

- ▶ $a < b$
- ▶ $f(x)$ is continuous over the interval $[a, b]$
- ▶ $f(x)$ is differentiable over the interval (a, b)
- ▶ $f'(x)$ is bounded over the interval (a, b) . That is, there exists a positive constant number F such that $|f'(x)| \leq F$ for all x in the interval (a, b) .

ERROR IN A SINGLE SLICE

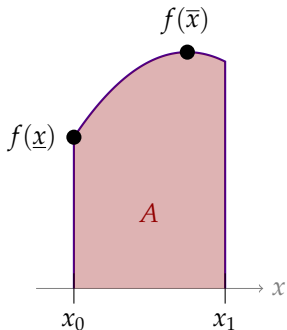
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► A is the actual area of the slice



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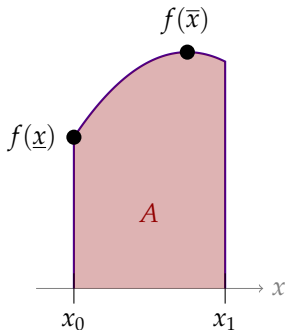
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- ▶ A is the actual area of the slice
- ▶ $f(\bar{x})$ and $f(\underline{x})$ are the largest and smallest function values over the slice

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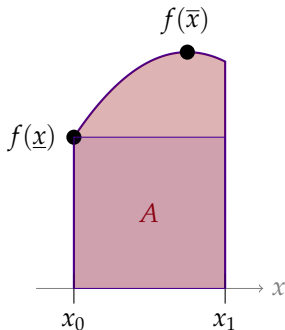
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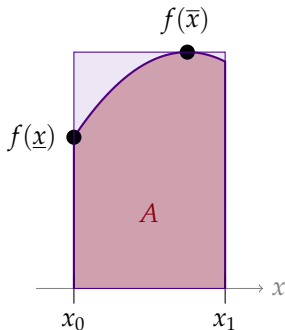
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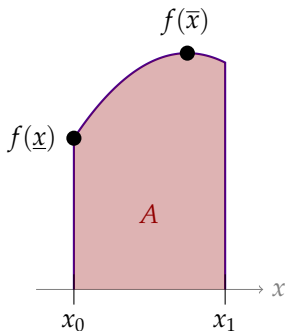
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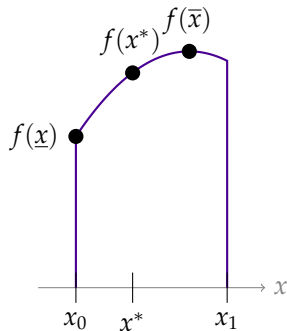
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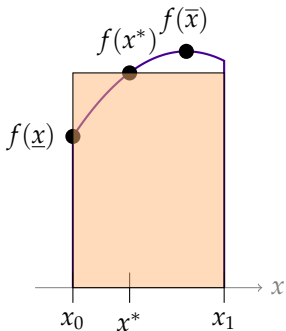
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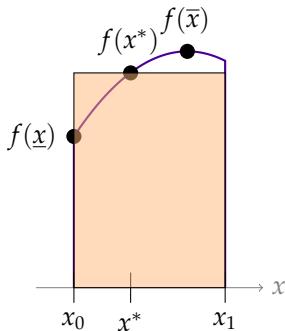
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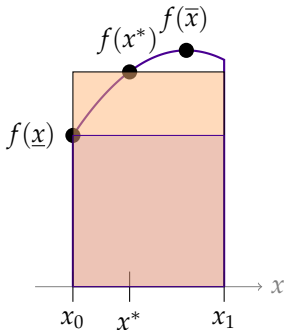


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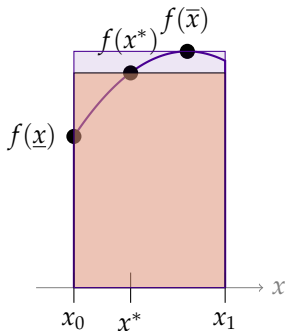
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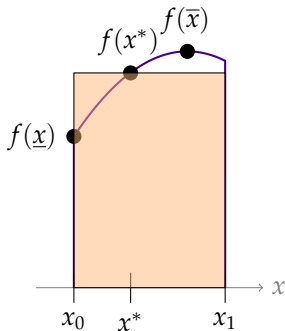
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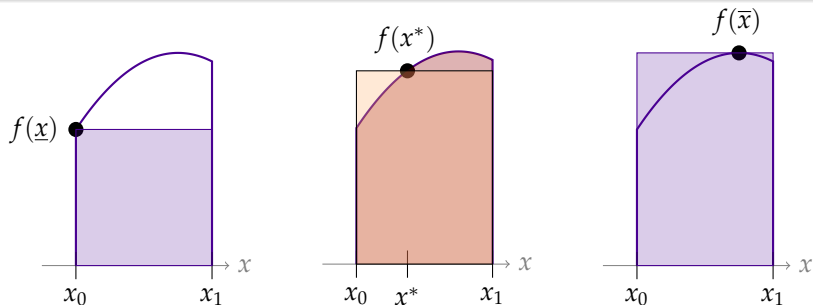
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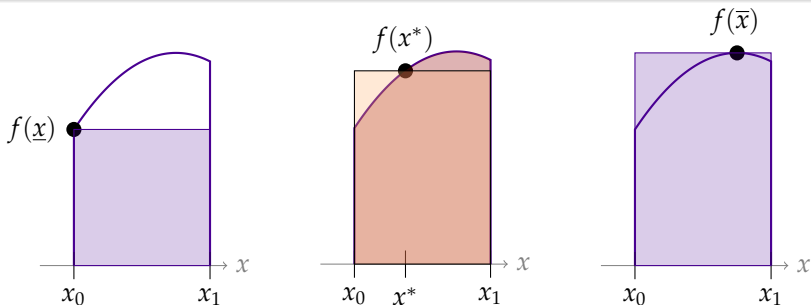
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ERROR IN A SINGLE SLICE



$$\begin{array}{rclclcl}
 f(\underline{x}) \cdot (x_1 - x_0) & \leq & A & \leq & f(\bar{x}) \cdot (x_1 - x_0) \\
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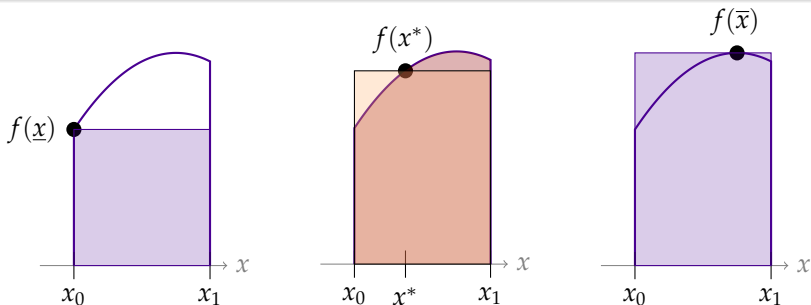
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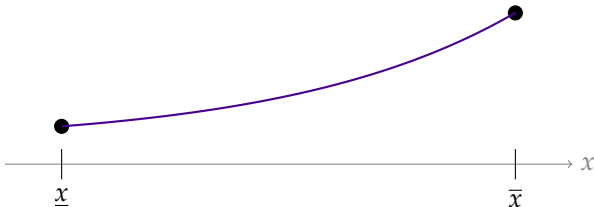
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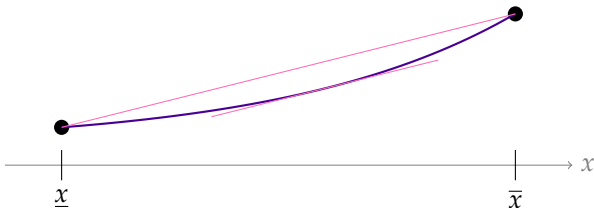
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$$f(\bar{x}) - f(\underline{x}) = f'(c) \cdot (\bar{x} - \underline{x})$$

Since $|f'(x)|$ is never larger than the positive constant F in (a, b) ,

$$|f(\bar{x}) - f(\underline{x})| \leq F \cdot |\bar{x} - \underline{x}| \leq F \cdot |x_1 - x_0|$$

ERROR IN A SINGLE SLICE

All together,

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$$\begin{aligned}\underbrace{|A - f(x^*) \cdot (x_1 - x_0)|}_{\text{error in slice}} &\leq [f(\bar{x}) - f(\underline{x})] \cdot (x_1 - x_0) \\ &\leq F \cdot |\bar{x} - \underline{x}| \cdot (x_1 - x_0) \\ &\leq F \cdot (x_1 - x_0) \cdot (x_1 - x_0)\end{aligned}$$

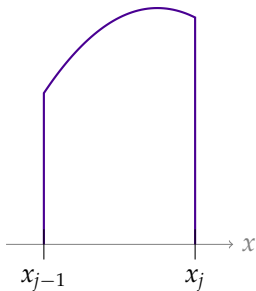
So,

$$\underbrace{|A - f(x^*) \cdot (x_1 - x_0)|}_{\text{error in slice}} \leq F \cdot (x_1 - x_0)^2$$

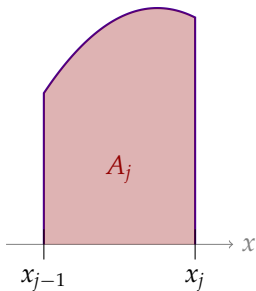
We have shown that the error on a **single** slice can't be worse than some amount.

Now let's consider adding up slices.

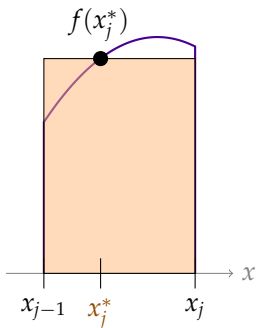
What we did for a single slice, we now do for all slices.
Updated notation for slice j :



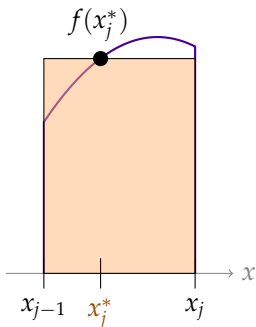
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Slice error bound:

$$\left| A_j - f(x_j^*) \cdot (x_j - x_{j-1}) \right| \leq F \cdot (x_j - x_{j-1})^2$$

(POSSIBLY IRREGULAR) PARTITIONS

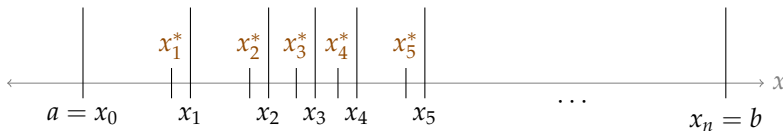
Consider partitioning the interval $[a, b]$ into n subintervals, not necessarily the same size. Let the points at the edges of the slices be $a = x_0, x_1, x_2, \dots, x_{n-1}, x_n = b$.



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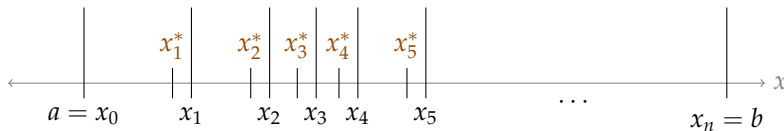
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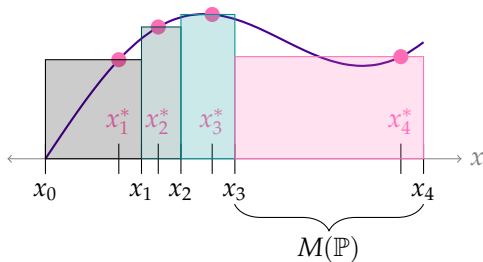
The approximation of $\int_a^b f(x) \, dx$ depends on how you choose your subintervals, and where you choose your sample points. Let

$$\mathbb{P} = (n, x_1, x_2, \dots, x_{n-1}, x_1^*, x_2^*, \dots, x_n^*)$$

denote these choices.

Let $\mathcal{I}(\mathbb{P})$ be the approximation that arises from $\mathbb{P}$:

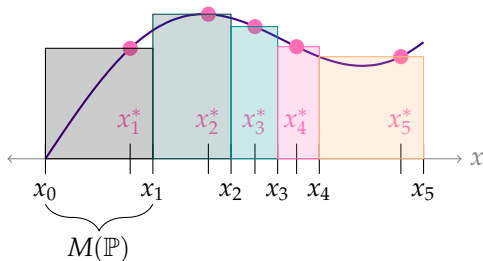
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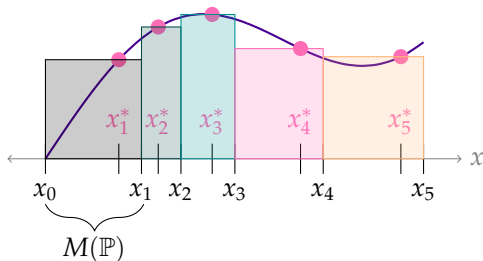
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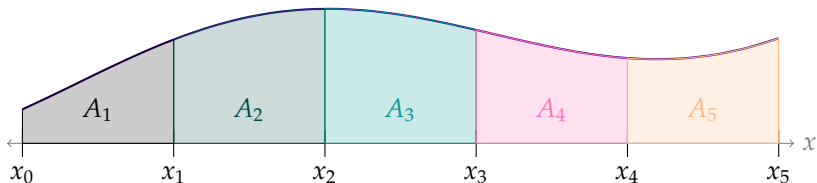
Let $M(\mathbb{P})$ be the maximum width of any subinterval.
If $M(\mathbb{P})$ is small, then *every* subinterval is small (narrow).

Define the integral as the limit

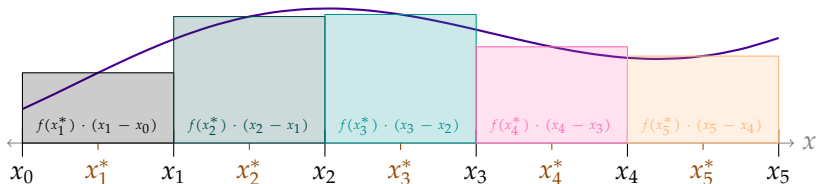
$$\int_a^b f(x) \, dx = \lim_{M(\mathbb{P}) \rightarrow 0} \mathcal{I}(\mathbb{P})$$

(Compare to our previous Riemann sum definition.)

We will show that the limit exists and is equal to the signed area under the curve.



Actual area: $\int_a^b f(x) \, dx = \sum_{i=1}^n A_i$



Approximation: $\mathcal{I}(\mathbb{P}) = \sum_{i=1}^n f(x_i^*) \cdot (x_i - x_{i-1})$

$$\begin{aligned}
 \underbrace{\left| \int_a^b f(x) \, dx - \mathcal{I}(\mathbb{P}) \right|}_{\text{overall error}} &= \left| \sum_{i=1}^n A_i - \sum_{i=1}^n f(x_i^*) \cdot (x_i - x_{i-1}) \right| \\
 &= \left| \sum_{i=1}^n [A_i - f(x_i^*) \cdot (x_i - x_{i-1})] \right|
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&= \sum_{i=1}^n F \cdot (x_i - x_{i-1}) \cdot (x_i - x_{i-1}) \\
&\leq \sum_{i=1}^n F \cdot M(\mathbb{P}) \cdot (x_i - x_{i-1})
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&\leq \sum_{i=1}^n F \cdot M(\mathbb{P}) \cdot (x_i - x_{i-1}) \\
&= F \cdot M(\mathbb{P}) \cdot \sum_{i=1}^n (x_i - x_{i-1}) \\
&= F \cdot M(\mathbb{P}) \cdot (b - a)
\end{aligned}$$

$$0 \leq \underbrace{\left| \int_a^b f(x) \, dx - \mathcal{I}(\mathbb{P}) \right|}_{\text{overall error}} \leq F \cdot M(\mathbb{P}) \cdot (b - a)$$

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$$\lim_{M(\mathbb{P}) \rightarrow 0} [F \cdot M(\mathbb{P}) \cdot (b - a)] = 0$$

So, by the squeeze theorem,

$$\lim_{M(\mathbb{P}) \rightarrow 0} \underbrace{\left| \int_a^b f(x) \, dx - \mathcal{I}(\mathbb{P}) \right|}_{\text{overall error}} = 0$$

That is,

$$\lim_{M(\mathbb{P}) \rightarrow 0} \mathcal{I}(\mathbb{P}) = \int_a^b f(x) \, dx$$

COMPARING DEFINITIONS

Here, we defined

$$\int_a^b f(x) \, dx = \lim_{M(\mathbb{P}) \rightarrow 0} \mathcal{I}(\mathbb{P})$$

for “nice” functions $f(x)$.

Originally, we used a slightly different definition:

Definition 1.1.9 (abridged)

For “nice” functions $f(x)$:

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_{i,n}^*) \cdot \frac{b-a}{n}$$

when the limit exists and takes the same value for all choices of the $x_{i,n}^*$ ’s.

COMPARING DEFINITIONS

We showed that **all** families of partitions “work,” as long as their largest subintervals shrink to length 0.

If all families of partitions “work,” then we might as well choose a simple one. The (arguably) simplest choices are regular partitions, cutting the interval $[a, b]$ into n subintervals of length $\frac{b-a}{n}$.